

K157 — Vol 1 Chapter 7 (Dynamics) Chapter-Grade Audit

Filed: 2026-05-22 Friday 09:18 EDT (Keeper, date-verified actual) Status: CONDITIONAL PASS framework-grade with one Cal #92(b) compliance flag for v1.0 path Mode: First chapter-grade textbook audit under new textbook-completion phase

Chapter under audit

notes/BST_Curriculum_Vol1_Ch7_Dynamics_v0_1.md — 218 lines, Lyra-authored, current version “v0.2 promotion filed Friday ~07:55 EDT” per Section 7.7. Front matter title claims “v0.3” — minor versioning inconsistency to clean up.

Scope: Schrödinger / Heisenberg / Path Integral dynamics on Bergman $H^2(D_{IV}^5)$ with $H_{\text{sub}} = \text{Casimir}$ on $L^2(D_{IV}^5; L_\lambda)$.

Verdict

CONDITIONAL PASS framework-grade. Chapter is structurally complete at framework-grade per Cal Mode 1 honest scope. ONE flag for v1.0 path closure.

What works (PASS items)

Methodology tier discipline — excellent

The chapter *correctly* labels itself “framework-grade not full chapter-grade” up front (note field + Section 7.0 + Section 7.4). The honest scope is visible to the reader, not buried.

Every theorem invocation in Section 7.6 (Theorem chain summary) has explicit status column: “Framework-complete” / “DERIVED” / “Framework” / “PENDING” with the OPEN multi-month item flagged. This is exemplary Calibration #19 discipline at chapter level.

Believability + Provability dual-axis — well-handled

Each major section (7.1, 7.2, 7.3) closes with explicit Believability + Provability paragraphs. The Believability framing is reader-accessible without overclaiming; the Provability framing cites specific theorems + toys.

Honest open scope (Section 7.4 “What’s NOT in this chapter”) — paper-grade

Section 7.4 explicitly enumerates four open items (explicit propagators, scattering amplitudes, time-ordered correlators, Wick’s theorem analog) as “multi-month pending Elie K52a Sessions 30+.” This is exactly the kind of honest framing Cal #92 referee-survival discipline requires.

Supporting K-audit anchors (Section 7.6b) — well-integrated

K108 (Hilbert space) + K111 (Casimir algebra) anchors are correctly cited as operational dependencies. The “dedicated K-audit on the dynamics framework” is flagged as Phase 3 Vol 1 work gated on operator-level closure — appropriate deferral.

Cross-chapter connections (Section 7.5) — clean

Ch 2, Ch 5, Ch 6, Ch 8, Ch 9, Ch 10 cross-references are at structural-dependency level, not content-overclaim level.

What needs work (CONDITIONAL items for v1.0)

Flag 1: Cal #92(b) header/body mismatch on T2457

Header claim (front matter v0.3 note): “Friday addition: T2457 Bergman = Feynman propagator identification”

Body framing (paragraph after that header + Section 7.3 + Section 7.5 Ch 9 cross-ref): “substrate reproducing kernel $K(z, \bar{w})$ plays propagator role in dynamics + path integral context”

These differ in tier level. Per Cal #92(b): section headers must match the tier-level framing within. “Bergman = Feynman propagator” is the overstated form; “Bergman plays propagator role” or “Bergman provides the structural role of the Feynman propagator” is the honest form.

Fix for v1.0: change front matter “Bergman = Feynman propagator identification” → “Bergman structural-role-of Feynman propagator identification” (or similar Cal #92(b)-compliant phrasing). The body content is already correctly framed; only the front-matter overclaim needs adjustment.

Flag 2: Version-number inconsistency

Front matter title: “v0.3 framework-grade narrative” Section 7.7 filing status: “v0.2 promotion filed Friday ~07:55 EDT”

These should agree. Either the title is wrong (v0.2) or Section 7.7 needs a “v0.3” entry documenting the Friday afternoon promotion absorbing T2457.

Fix for v1.0: Lyra reconciles to consistent version number.

F1-F4 (chapter-grade)

- F1 framework-grade structural completeness + honest scope + methodology tier discipline: 3.9/4
- F2 theorem chain cross-paths (T2438 + T2435 + T2428 + T2430 + T2429 + T2437 + T2405 + Elie S29): 3.95/4
- F3 cross-lane (Lyra primary + Elie S29 H_{sub} + Cal #90 framework PASS at v0.2): 3.85/4
- F4 chapter-grade falsifier (operator-level open + Wick analog open): explicitly enumerated: 3.9/4

F1-F4: 15.6/16 = 3.90/4 STRONG — chapter-grade framework PASS

B1-B4 (chapter-grade)

- B1 Schrödinger / Heisenberg / Path-Integral framework on Bergman $H^2(D_{IV}^5)$: 3.9/4
- B2 T2438 + Elie S29 + Ch 2 + Ch 5 + Ch 6 cross-anchor: 3.95/4
- B3 alt-substrate dynamics framework comparison: 3.7/4 (multi-month gate)
- B4 curriculum integration: 3.9/4 (Ch 7 fits between Ch 6 operator zoo and Ch 8 gauge theory + Ch 9 scattering + Ch 10 renormalization)

B1-B4: 15.45/16 = 3.86/4 STRONG

Cross-volume consistency check (Keeper sweep)

Verified no contradictions with: - Vol 0 Ch 9 Conservation Laws (Keeper) — Ch 7 conservation laws (energy, K-type, Casimir) consistent with Vol 0 framework - Vol 1 Ch 2 Substrate Hilbert Space — $H^2(D_{IV}^5)$ consistent across chapters - Vol 1 Ch 5 Casimir Algebra — H_{sub} = Casimir consistent - Vol 1 Ch 6 Operator Zoo — 6/6 operators referenced consistently

No cross-volume contradictions found.

Path to v1.0 for Vol 1 Ch 7

For Ch 7 to reach v1.0 venue-publishable state:

1. Cal cold-read PASS at v0.3 (currently v0.2 PASS, v0.3 not yet PASS'd) — PRIORITY 1
2. Fix Flag 1: Cal #92(b) header/body mismatch on T2457
3. Fix Flag 2: version-number consistency
4. Reader-grade polish: substrate-tick $GF(128)^k \rightarrow$ continuum $H^2(D_{IV}^5)$ projection visualization (Section 7.7 already flagged)
5. Operator-level closure: explicit propagators + scattering + correlators (Elie K52a Sessions 30+ multi-month, NOT blocker for v1.0 at framework-grade per Cal Mode 1 honest scope)

Estimate: 1-2 days for Lyra-side fixes; Cal cold-read can happen in parallel. Operator-level closure is multi-month but doesn't gate v1.0 framework-grade chapter — the chapter explicitly closes that scope question via Section 7.4.

K157 status

K157 CONDITIONAL PASS framework-grade Vol 1 Ch 7 Dynamics.

Compliance with Calibration #19 standing rule: EXCELLENT (honest scope visible, methodology tiers labeled, open items enumerated). Two minor v1.0 path items flagged (Cal #92(b) header fix + version consistency).

This is the first chapter-grade textbook audit under the textbook-completion phase. The template demonstrated here applies to subsequent Vol 1 / Vol 2 chapters.

— Keeper, K157 Chapter-Grade Audit filed Friday 09:18 EDT (date-verified actual)